

E-trade Management System Architecture

Oleg Pursky
department of computer science
Kyiv national university of trade and
economics
Kyiv, Ukraine
e-mail: Pursky_O@ukr.net

Pavlo Demidov
department of computer science
Kyiv national university of trade and
economics
Kyiv, Ukraine
e-mail: dempg@meta.ua

Anna Selivanova
department of computer science
Kyiv national university of trade and
economics
Kyiv, Ukraine
e-mail: a.selivanova@knute.edu.ua

Volodymyr Kulazhenko
department of cybernetics and system
analysis
Kyiv national university of trade and
economics
Kyiv, Ukraine
e-mail: v.kulazhenko@knute.edu.ua

Oleksander Kharchenko
department of software engineering and
cyber security
Kyiv national university of trade and
economics
Kyiv, Ukraine
e-mail: a.kharchenko@knute.edu.ua

Abstract—In this work we present the model for construction of e-trade management system architecture which consists of two basic blocks: the landing Web-pages management unit of the online store is responsible for interacting with the user, receiving orders and collecting primary information and Web-enterprise management unit, responsible for processing orders, their delivery, and all enterprise's business processes realization. We have focused our attention to developing an architecture model for e-trade management system. A service-oriented architecture (SOA) using enterprise service bus (ESB) concept is offered for architectural implementation. In developed e-trade management system architecture we used the method of integrating complex information systems within the information environment based on SOA and ESB technologies, which represents an independent, integrative subsystem for relatively integrated information systems. Typical information systems that belong to the e-trade system are used as services. These web-based services should be developed and used to enhance ESB efficiency. Such web-services implement SOAP and use WSDL (Web Services Description Language), as well as UDDI specification (Universal Description, Discovery and Integration). At that, each typical information system is simultaneously data supplier and data consumer. Each information system handles its own data class, while being able to both transmit and receive, adjust and modify data according to operational information from other information systems within main business processes.

Keywords—*e-trade, management system, SOA, ESB, architecture model.*

I. INTRODUCTION

Within last ten years, modern information technologies have transformed Internet in the developed infrastructure which covers all basic information centers, scientific and legal information databases, world libraries, and many state and commercial organizations. At present, the Internet can be seen as a huge electronic market, which is potentially able to reach practically all of the world's population [1]. For this reason, manufacturers of software and hardware, trading and financial organizations are actively developing various types and methods of doing business in the global network, the so-called electronic commerce.

As a result, electronic trading has become a very profitable form of relationship with the buyer who is developing, not seeking to replace it with other forms of trade contacts and connections. Economic properties and peculiarities that have emerged in the process of becoming

Internet commerce have not only provided it with the possibility of a competitive global existence in the world of modern global business, but also created the prerequisites for quite optimistic forecasts of its future [2].

When organizing electronic commerce it is impossible to do without the specialized software - systems of e-commerce. Today, enterprises use the different information systems that have different characteristics in terms of security, scalability, performance, etc., and have different costs. With a large variety of proposed solutions, it is difficult for user to choose the one that would ensure effective work in the field of e-commerce.

Use of Internet technologies in everyday business life a few years ago became the norm for Western companies. Global network it is used not only for messaging and for accessing information resources, but also for conducting specific commercial operations that constitute the essence of e-commerce. Development of such technology helps to reduce costs and accelerate all business-processes that, as a result, increasing the profitability of all activities company. Today, for any modern enterprise the high-technologies and quality of implementation information technology in the significant degree determines its success in the market. The competitiveness of goods and services directly depends on the level of information technology application. Companies that manufacture of equipment and software first began to use the Internet in the field of marketing and trade. An example of successful development many of them became the factor of attractiveness for firms operating in the other areas [3].

The effectiveness of the electronic trading markets operation depends on the number of potential consumers of products and services presented on the market. An electronic trading environment cannot be formed without a wide range of consumers. Their presence is determined by the level of development and the degree of implementation of information and communication facilities in all spheres of socio-economic activity. Without providing users with Internet access, it's difficult to talk about the availability of an electronic trading environment. The presence of vendors using computer network resources for the sale of goods and services is a proactive factor in the electronic trading environment formation, since sellers are interested in making profits through the use of new, electronic trading mechanisms. At the same time, in order to have an electronic

trading environment, sellers have to inform potential buyers in virtual space, while they bear the appropriate promotion expenses and market entry [4]. The cost of electronic trading market entering, it is by definition the financial and material costs to ensure the operation of the Internet store (website creation, delivery of goods organization, electronic payment systems connection, etc.). Promotion expenses in electronic trading are related to the creation of mechanisms for influencing consumers' preferences, by forming the priority consumer goods value. Since there is little or no cost of obtaining information about a particular product or service in an online environment, the role of informative advertising in electronic commerce is significantly restricted, that is, the overwhelming part of advertising is a manipulative advertising, which increases the readiness of potential buyers to buy goods and services [2]. Promotional expenses in electronic trading primarily relate to the arrangement and placement on various electronic resources (own websites and other areas providing advertising) the banners about services and products and various incentive measures. Thus, potential buyers will get information about sellers and can interact with sellers through information and communication tools.

Electronic trading markets operation has caused significant changes in the modern economy. The basic relations and business models have changed: from the manufacturer's domination, and then to the intermediary - to the leading role of the consumer; From mass production - to manufacturing products to order; From the value of material resources - to the value of information. At the same time, electronic trading markets have not changed the basic concepts of market relations, but have identified new mechanisms for their implementation.

The electronic trading market is a prevailing electronic trading environment and a set of its participants, products, services and processes of their interaction, characterized by certain regularities in the context of a developed network of information and communication features. An important circumstance that ensures the electronic trading markets operation is the availability of a fully-fledged electronic trading environment. The electronic trading environment is formed in the presence and direct interaction of such factors: modern information and communication tools for e-commerce, a wide range of consumers with full access to information and communication facilities (with access to the computer network Internet), sellers, using information and communicative means for the sale of goods and services. Information and communicative tools for electronic trading are the modern information technologies and systems that provide effective implementation of procedures for reading, processing, storing and transfer of information resources based on computer networks [2].

The development of models and methods of constructing management information systems for e-trade enterprises remains relevant due to the emergence of new mechanisms and tools for electronic commerce. Therefore, the considerable interest of scientists in the problem of the building effective management systems for the economic activity of e-trading enterprises is fully justified. The purpose of this work was to develop an architecture model for e-trade (ET) management system.

II. RESULTS AND DISCUSSION

From the functional point of view, all processes in electronic trading are divided into two blocks: 1) The landing Web-pages management unit (LPMU) of the Internet shop is responsible for interacting with the user, receiving orders and collecting primary information; 2) Web-enterprise management unit (WEMU), responsible for processing orders, their delivery, and all enterprise's business processes realization.

The landing Web-pages management unit (LPMU). The work of each Web Store (WS) involves the preparation of a range of products (module for managing the assortment of goods and services) and store management (WS management module). These modules refer to the WEMU block and will be considered further in a detailed analysis of this contour. On the basis of information about the range (module for managing the assortment of goods and services) and the business rules of a particular store (WS management module), data is generated for its showcases (shop window management module).

The user raises an order based on information about the range of goods and services coming to the shop window from the assortment management module and the terms of the purchase of goods (WS management module). In order to realize the optimal purchase process, the potential buyer needs information not only about the range of goods and services, but also the detailed description and graphic representations of the products, information about their manufacturers, the specifics of the services and other additional information. Informational support for goods and services requires the availability of an appropriate information and analytical module for managing information about goods and services. Its functions include: gathering information about goods and services; systematization, structuring, updating and processing of information; data representation formats harmonization; tracking and making appropriate changes; providing complete information on the characteristics of goods and services.

Consumer demand for goods and services formation is the task of the company's marketing and advertising departments, which use various applications in their work that implement the functions of the module for managing the advertising of goods and services. Its main functions are: demand for goods and services formation; advertising and marketing information preparation; consumer reviews monitoring; statistics monitoring; effectiveness of advertising and marketing companies determination. In order to accomplish the described tasks, the advertising management module uses the information coming from the management modules for the range of goods and services and the WS management. Marketing and promotions have an impact on the target audience in order to obtain orders from the online store. When ordering, the user becomes a customer of the online store. The module for managing the content of WS provides a solution to the tasks associated with the placement and submission of information in relation to the specific display of a specific WS, determined by the management module of WS. The consumer management module provides for: informing them; personal interaction with clients; personalization of information representation in accordance with the interests of each client; customer loyalty management.

The functional module for managing the shop window of WS is the resulting module for all processes related to the formation of this window and its information content. At a particular shop window of WS the consumer carries out actions on the selection of the goods, the review of information about it, the terms of purchase and the order execution. The process of ordering consists of the following steps: a showcase identifies the consumer; the consumer indicates the delivery address, chooses the terms of payment and delivery and confirms his or her choice; the showcase forms the buyer's order and provides related information. Further in the system of electronic trading under the "order" is understood the set of entities: user, order, terms of service.

Informational entities are associated with a specific WS and its shop window, on which the order has been made. The essence of "goods", "consumers" and "orders", being the part of the information environment of the Web-enterprise, relate to a specific WS, which creates the possibility of conducting targeted work with different groups of buyers and implementing additional mechanisms of economic and manipulative impact on consumers and work with assortment of goods and services.

It should be noted that an important advantage of working with the network of showcases, which is already foreseen at the stage of constructing a architecture model, is the possibility of cross-interaction of showcases. Interaction of Internet shop windows involves the mechanism of unified registration and authorization of consumers, interaction with third-party payment systems and other joint actions. In particular, the user's order may be taken into account in the rules of the display of another showcases in such a way as to provide the user with advantages in choosing and terms of purchase of goods or services. For example, if a user bought a computer in one showcase, then another showcase, as a special offer, will offer textbooks on the use of computers.

Web-enterprise management unit (WEMU). The work of WS begins with the formation of an assortment of goods and services. This task of module for managing the range of goods and services based on information from suppliers and manufacturers of goods and services. Main functions of the module of assortment of goods and services management: monitoring of the state of the markets; tracking the load of warehouse suppliers; monitoring the prices for goods and services; decision-making on price formation; making decisions on the assortment formation; updating and tracking data on the range of goods and services specific to WS. WS management (WS management module) is determined by the requirements for its operation from side of business. These requirements are a characteristic feature of a specific WS and are presented by the conditions of assortment formation, terms of payment and delivery of goods, terms of conducting trade, marketing and promotional acts, content filling of showcases.

After the receipt of the buyer's order to the order and service module, the verification of incoming information and tracking the status of buyers orders execution is carried out. At this stage, there is a presence or absence of contradictions in the possibilities of the WS and the requirements of the buyer, the conditions of execution of the order and service of the buyer are determined, the conditions of work with the goods are determined, the terms are calculated and the methods of execution of the order are specified, with the

toolkit of the service of orders and services interacting managers of WS customer service.

Financial management (financial management module) provides interaction with accounting or banks to provide trading operations financially. The financial management module also implements the functions of interaction with payment systems, monitoring and allocation of financial resources on accounts and directly interacts with the modules of resource management and supply management.

The operation of all modules takes place taking into account the constantly operation information and business processes for eliminating the shortage of goods and services, processing orders, accounting and distribution of resources, collecting and analysing the current status of order processing and statistical data on relationships with counteragents for the adoption of the following management decisions. The stock accounting for order maintenance and order batching is carried out at the expense of the module functions ordering implementation on the basis of data coming from the resource management module, which manages the resources of the Web-enterprise and performs the functions of accounting for material resources. When the goods are purchased and distributed by orders, and the orders have the necessary documentation, it is necessary to ensure the delivery of goods to customers. The order delivery management module is the final stage in the process of customer order processing and is consistent with the functions of the service delivering orders to customers. The delivery service receives information and resources from the process (service orders and services - supply management - financial management - resource management - complete orders) and transmits them for execution. Successful delivery of goods to the buyer is accompanied by the receipt of financial resources from him or her in a cyclic process (the range of goods and services management - ... - orders delivery management).

Figure 1 displays developed architecture model for e-trade management system. Data exchange between services (local information sestems) is held using corporate service bus according to the following scheme: A service supplier registers his services in a register, and a customer refers to the service register via enterprise service bus. In order to use the service, interface agreement abidance is vital. Service access interface should not depend on software application platform. ESB and SOA application provides services scalability, possibility of their enhancement and modernizing. Service provider and customer are not connected with the cross-platform interaction means, they are fully independent. Since interface, data exchange methods and message formats are platform-independent, a flexible and scalable informational environment is formed with a potential to handle events independently in real-time mode, within specific information sestems, without impact on functioning mechanisms in general and independent on operational conditions. SOA application by means of using ESB concept allows to implement "bus" integration pattern within WEMU loop. It means a high level of informational integration for ET informational and business processes. Application of SOA and ESB improves WEMU loop flexibility and decreases information systems dependencies within loop structure, providing a high level of informational connection between information systems inside the loop [5]. WEMU loop architectural organization is based on SOA and

ESB whose main participants are service suppliers and service register. Typical information systems (IS) belonging to WEMU loop are used as typical services [6]. At that, each typical information system is simultaneously data supplier

and data consumer. It is important that data are available for online operation.

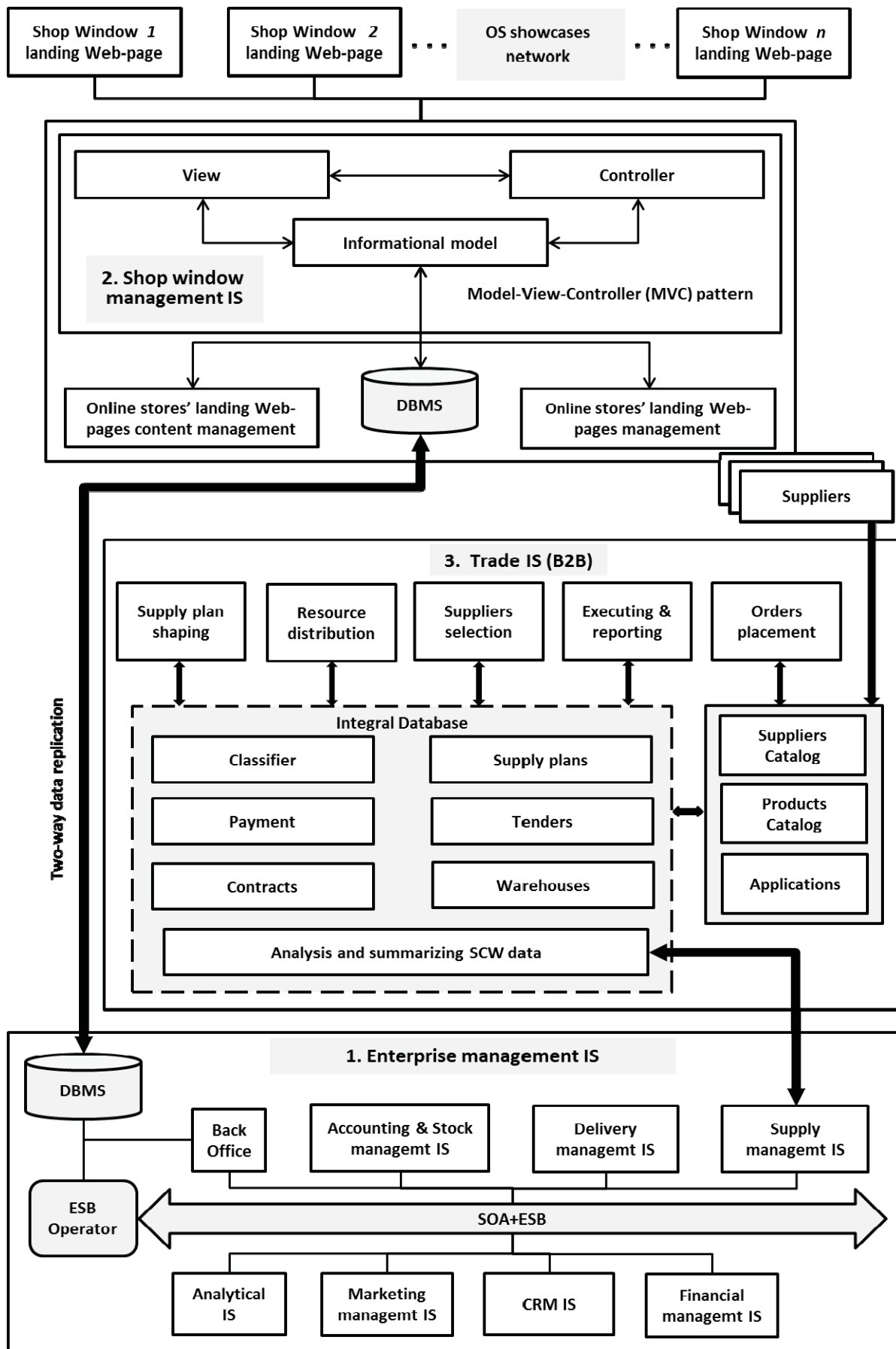


Fig. 1. Architecture model for e-trade management system. OS – online store, IS – information system

Each information system handles its own data class, while being able to both transmit and receive, adjust and modify data according to operational information from other information systems and transmit data according to respective procedures, verifying their actuality at each stage of integrated information exchange within main business process. In this way, ET business process apparatus and its formal logic scope are held to independent abstracted level that significantly improves management staff capabilities in business process modification. Using SOA and ESB concepts implemented in software complex allows acquiring practical architectural solution. Data transferred between information systems are independently packaged using Simple Object Access Protocol (SOAP) and XML as data exchange protocol and format [7]. Packages are entities (informational objects) circulated between information systems, such as orders, delivery notes, supply inquiries, etc. Service register is a separate information systems, enterprise service bus operator. It is important to notice that web-based services should be developed and used to enhance ESB efficiency [8]. Such web-services will implement SOAP and use WSDL (Web Services Description Language), as well as UDDI specification [9] (Universal Description, Discovery and Integration). These means make it possible to modify information systems list as necessary, without modifying each of the services. It allows also to flexibly vary typical information systems set that belong to informational environment, which is important enough for ET enterprise development [10]. Unlike the "top-down" design method, in which the information systems development process begins with the global formalization of processes and data and ends with detailed design, in this study we used the modern alternative approach to designing the information systems is called "bottom-out", it aims to find methods for creating systems with elements of self-management. For such systems IBM proposed the term "autonomic computing", in other editions also the term autonomous systems [11]. An example of such an approach can be the Internet in which the connection of resources is implemented by integrating of independent information systems - Internet services (DNS, bind, routing, etc.), which provide network management with the independence of all nodes [12]. In this case, technological implementation simultaneously provides communication and integration of independent information systems (integrating circuit, fig. 1) and their management (management loop, fig. 1), which provides true vertical integration. But such "vertical" does not bind of these information systems, but bind the two levels of representation - the management and the integration. Moreover, there is an additional opportunity to integrate other typical information systems over time. For example, integration BPM into the structure of e-trade enterprise and implementation modern strategic management standards for a large company based on the management and business processes optimization.

III. CONCLUSIONS

In developed e-trade management system architecture we used the method of integrating complex information systems within the information environment based on SOA and ESB technologies, which represents an independent, integrative subsystem for relatively integrated information systems. The architecture of SOA and ESB provides the construction of a flexible and high degree of integrated environment.

However, in this modern and actively implemented corporation's method there is no solution to the management problem. In this regard, it is necessary to improve the proposed model of integration of large number of basic information systems in common information environment with efficiency close to the level of vertical integration, but with the same flexibility and compliance with the basic requirements for automation of information systems. The corporate environment requires rigorous management, driven by the company's desire to control each process at any given time, because the very high level of control provides high controllability and the ability to increase profits due to timely management decisions. In most corporate information systems, this type of management is implemented on the basis of the central management program. Accordingly, in e-trade management system, where the number and complexity of the integrated elements is significantly higher, it is necessary to ensure the same quality of management as in the corporate information systems with software control.

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